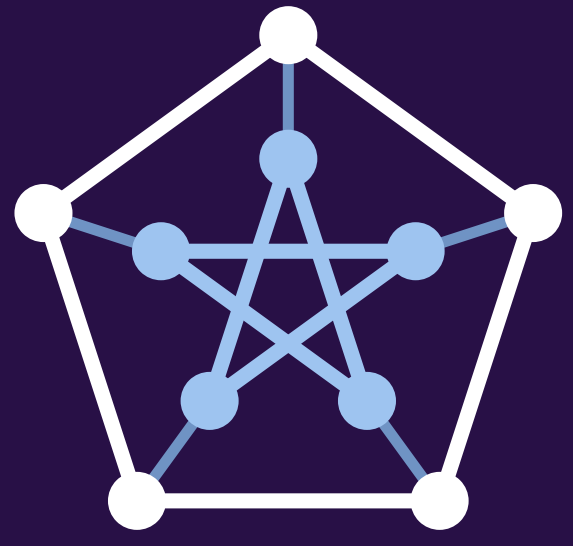




Ramanujan Graphs and Zero Forcing in the Generalized Petersen Family

A certified classification of the graphs satisfying the Riemann Hypothesis, and the exact zero forcing number for every large n

Aryan Padarthy · Allen High School · Allen, Texas, USA · Mathematics

Introduction

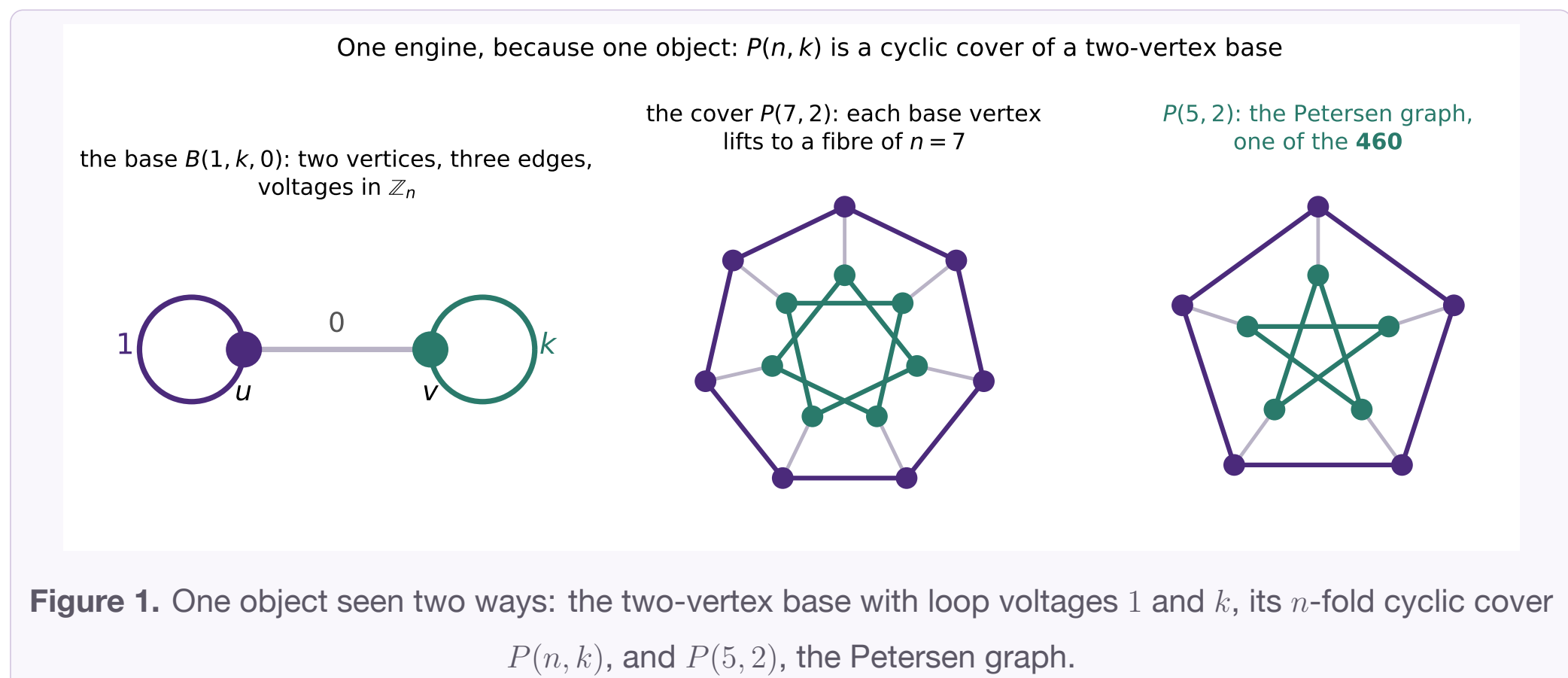
The **generalized Petersen graph** $P(n, k)$, $1 \leq k < n/2$, has outer vertices u_i and inner vertices v_i ($i \in \mathbb{Z}_n$) with edges $u_i u_{i+1}$, $v_i v_{i+k}$, $u_i v_i$: cubic, on $2n$ vertices, with the rotation $\rho : i \mapsto i + 1$ as an automorphism of order n . This board asks two questions about it, one about its fixed adjacency matrix and one about every matrix with its pattern.

What was known. Gera–Stănică [2] computed every eigenvalue of $P(n, k)$ in closed form but never asked which members are Ramanujan; Droll and Le–Sander [6] classified the Ramanujan members of other families (finitely many each). Rashidi et al. [4] proved $Z(P(n, k)) \leq 2k + 2$ and claimed $Z(P(n, 3)) = 8$ for $n \geq 12$; Krishnan [5] exhibited $Z(P(12, 3)) = 7$, tabulated $n \leq 20$, and conjectured $Z(P(n, 3)) = 8$ for all $n \geq 13$, lacking a lower bound valid for every large n . Nothing was known for $k \geq 4$.

Research questions. **Q1.** For which (n, k) is $P(n, k)$ Ramanujan, decided exactly and provably completely?

Q2. Is $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all large n , and from which n on?

Framework



Ramanujan graphs and the graph Riemann Hypothesis. A cubic graph is *Ramanujan* if every eigenvalue other than 3 (and -3 when bipartite) satisfies $|\lambda| \leq 2\sqrt{2}$; by Alon–Boppana this is the best spectral gap an infinite family can have. Ihara's zeta function $\zeta_X(u) = \prod_{[C]} (1 - u^{\ell(C)})^{-1}$ over prime cycles satisfies $\zeta_X(u)^{-1} = (1 - u^2)^{|E| - |V|} \det(I - uA + 2u^2I)$, so its poles are the roots of $2u^2 - \lambda u + 1 = 0$, one quadratic per eigenvalue λ . With $u = 2^{-s}$ they lie on $\operatorname{Re} s = \frac{1}{2}$ iff $|u| = 1/\sqrt{2}$ iff the roots are complex conjugates iff $\lambda^2 \leq 8$ [1].

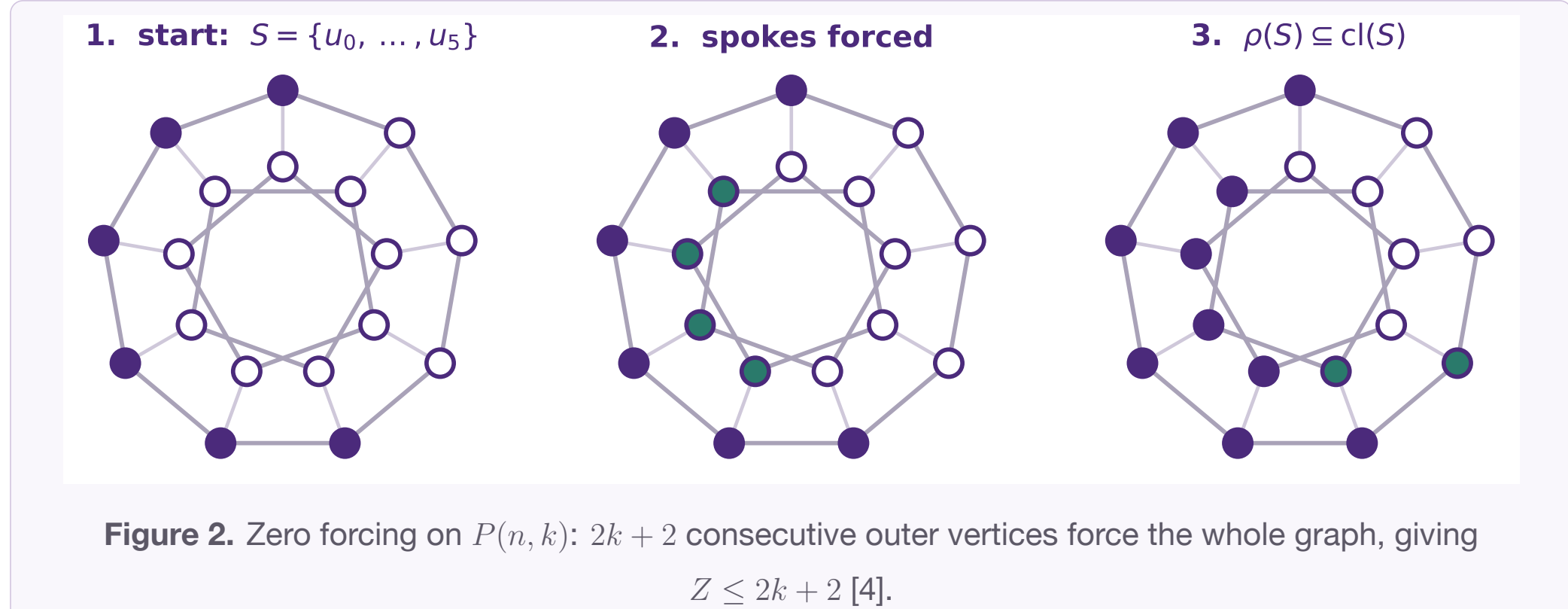
Proposition (Ihara; Terras [1]). A cubic graph satisfies the Riemann Hypothesis for ζ_X if and only if it is Ramanujan. The analogy with the classical RH lives in the zeta function, not in the difficulty: the graph statement is a finite eigenvalue check.

Zero forcing and maximum nullity. Colour a set $S \subseteq V$ blue. If a blue vertex has exactly one white neighbour, that neighbour turns blue. S is a *zero forcing set* if everything ends blue; $Z(G)$ is the least $|S|$. For a symmetric matrix A with $A_{xy} \neq 0 \iff xy \in E$ ($x \neq y$) and free diagonal, $M(G)$ is the largest possible null A .

Proposition (AIM group [3]). null $A \leq Z(G)$ for every matrix with the pattern of G ; hence $M(G) \leq Z(G)$.

Proof sketch. If $Ax = 0$ and x vanishes on the blue set, row u of $Ax = 0$ reduces to $A_{uw}x_w = 0$ for the unique white neighbour w , and $A_{uw} \neq 0$ forces $x_w = 0$. A kernel vector vanishing on a forcing set is 0, so $\ker A$ is determined by its values on S . $\square$

A matrix of nullity $2k + 2$ is therefore a *certificate* for a lower bound on Z , checkable by elimination.



The spectrum on a grid. ρ commutes with A , so the Fourier transform over $\mathbb{Z}_n$ splits A into 2×2 blocks $M_j = \begin{pmatrix} \alpha_j & 1 \\ 1 & \beta_j \end{pmatrix}$, $\alpha_j = 2 \cos \frac{2\pi j}{n}$, $\beta_j = 2 \cos \frac{2\pi jk}{n}$, with eigenvalues $\frac{\alpha_j + \beta_j}{2} \pm \sqrt{\left(\frac{\alpha_j - \beta_j}{2}\right)^2 + 1}$: for fixed k , two fixed curves in $t = 2\pi j/n$ sampled at n points. Both halves of the project live on this grid of n -th roots of unity.

Findings

Part I — which $P(n, k)$ satisfy the Riemann Hypothesis

Theorem 1 (polynomial criterion). Let $u = \cos \frac{2\pi j}{n}$ and T_k the Chebyshev polynomial, and put

$$D_k(u) = (7 + 4uT_k(u))^2 - 8(2u + 2T_k(u))^2 \in \mathbb{Z}[u], \quad \deg D_k = 2k + 2.$$

Then $P(n, k)$ is Ramanujan iff $D_k(u_j) \geq 0$ for every nontrivial j .

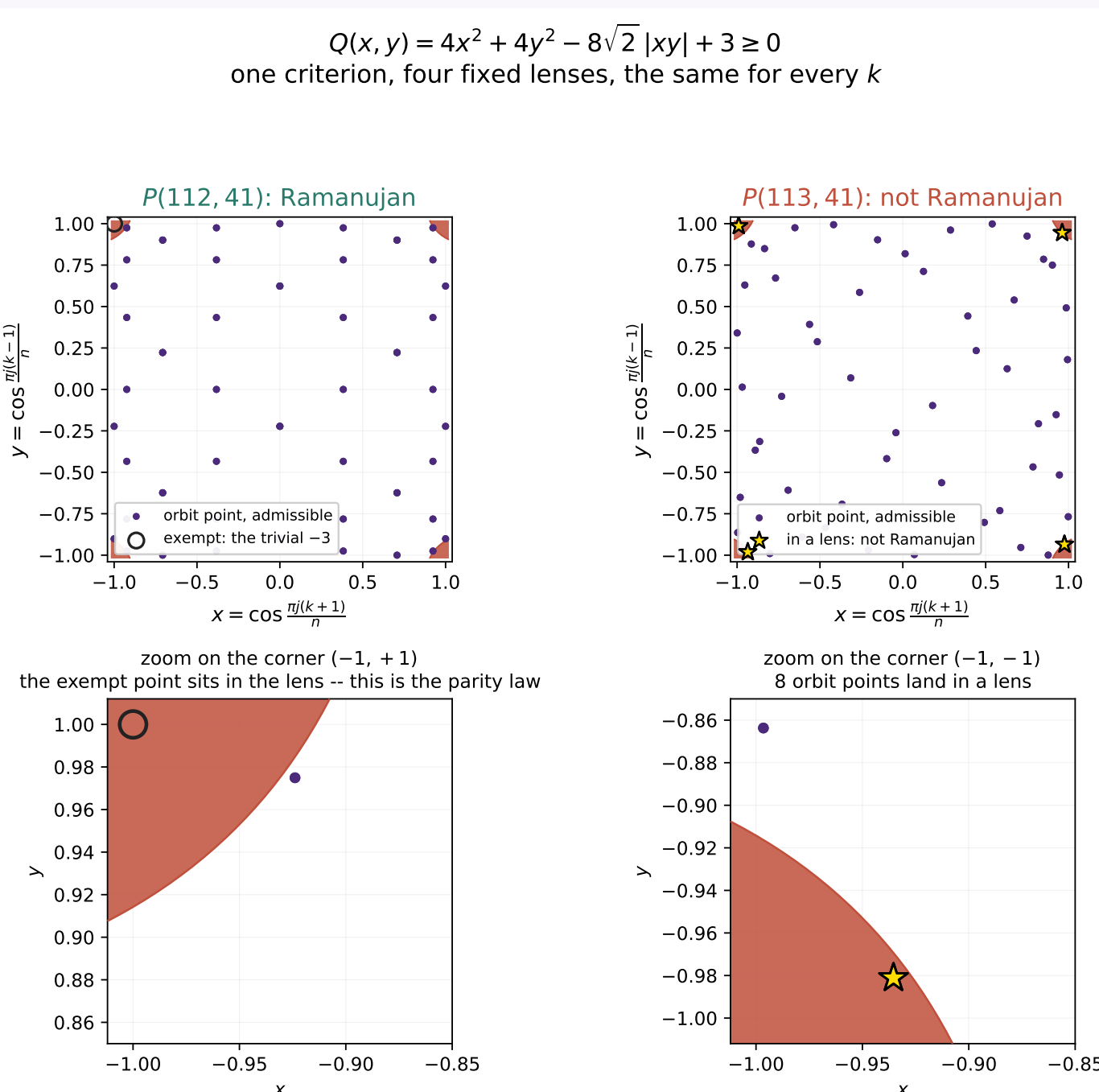
Proof sketch. Write $A = \alpha + \beta$, $P = \alpha\beta$. The condition $\lambda_+ \leq 2\sqrt{2}$ is $\sqrt{(\alpha - \beta)^2 + 4} \leq 4\sqrt{2} - A$; the right side is $\geq 4\sqrt{2} - 4 > 0$, so squaring is an equivalence, and with $(\alpha - \beta)^2 = A^2 - 4P$ it becomes $2\sqrt{2}A - P \leq 7$. The mirror condition gives $2\sqrt{2}|A| \leq 7 + P$; since $7 + P \geq 3 > 0$ a second squaring is again an equivalence: $8A^2 \leq (7 + P)^2$. Substitute $\beta = 2T_k(u)$. $\square$

Theorem 2 (corner criterion). With $x = \cos \frac{\pi j(k+1)}{n}$ and $y = \cos \frac{\pi j(k-1)}{n}$, $P(n, k)$ is Ramanujan iff, at every nontrivial j ,

$$Q(x, y) = 4x^2 + 4y^2 - 8\sqrt{2}|xy| + 3 \geq 0.$$

Q does not depend on k , and $\{Q < 0\}$ is four congruent corner pieces covering 0.43% of the square.

Proof sketch. $D_k = f_- f_+$ with $f_{\pm} = 7 + P \pm 2\sqrt{2}A$ and f_- or $f_+ \geq 6$, so only $\min f_{\pm} \geq 0$ matters. The identities $\cos t + \cos kt = 2xy$ and $\cos t \cos kt = x^2 + y^2 - 1$ give $A = 4xy$, $P = 4x^2 + 4y^2 - 4$.



Theorem 3 (finiteness, three ways). (a) $D_k(1) = -7$ for every k , so $n \leq B_k = 2\pi / \arccos u_k$, u_k the largest root of D_k below 1; $B_2, B_3, B_4 = 23, 32, 42$, all sharp. (b) Uniformly in k , and (c) for every k at once:

$$n \leq \pi(2 + \sqrt{2})\sqrt{k^2 + 1}, \quad n \geq 231 \rightarrow \text{not Ramanujan.}$$

(d) Parity: $D_k(-1) = -7$ for odd k , and odd n is then cut at exactly half the constant; even n is exempt, its sample being the trivial -3 of a bipartite graph.

Proof sketch. (b) With $a = 1 - \cos t$, $b = 1 - \cos kt$, $s = a + b$, failure reads $(8\sqrt{2} - 4) - (4\sqrt{2} - 4)s - 4ab > 7$; AM–GM $4ab \leq s^2$ and the identity $s^2 + (4\sqrt{2} - 4)s = 8\sqrt{2} - 11$ at $s = \tau := 3 - 2\sqrt{2}$ show $s < \tau$ suffices; $1 - \cos \theta \leq \theta^2/2$ gives $s \leq (1 + k^2)t^2/2$ at $t = 2\pi/n$. (c) Failure follows from $\|j/n\|^2 + \|jk/n\|^2 < \tau/2\pi^2$ for some j . Dirichlet's theorem with $J = \lfloor \sqrt{n} \rfloor$ gives $j \leq \sqrt{n}$ with $\|jk/n\| < 1/\sqrt{n}$, so the left side is $< 2/n$, and $2/n < \tau/2\pi^2$ iff $n > 4\pi^2/\tau = 230.1$. $\square$

Theorem 4 (complete classification). Exactly **460** pairs (n, k) satisfy the Riemann Hypothesis, 324 graphs up to isomorphism, out of 13110 cases with $n < 231$. The largest is $n = 112$ ($k = 41$); none has $k \geq 45$.

The list follows from a structural statement: $P(n, k)$ is Ramanujan iff no divisor m of n is *bad* for k mod m (locality), where badness is witnessed by a single j with $Q < 0$ and $m \leq 231$. So the classification is a theorem together with a finite table of bad levels, each entry decided exactly in $\mathbb{Z}[\zeta_m]$. For small k no table is needed:

Theorem 5 (closed form for $k \leq 9$). For $k \in \{1, 2, 3, 4, 5, 7, 9\}$ the negative set of D_k on $[-1, 1]$ is the end band $(u_k, 1]$ alone (even k) or $[-1, -u_k) \cup (u_k, 1]$ (odd k), by exact root isolation of one integer polynomial; hence $P(n, k)$ is Ramanujan iff $2k < n \leq B_k$, and for odd k additionally $n \leq B_k/2$ when n is odd. For $k = 6, 8$ and $k \geq 10$ an interior band appears and the list has gaps.

Part II — the exact zero forcing number

Theorem 6 (the ceiling). For every matrix A with the pattern of $P(n, k)$, symmetric or not,

$$\text{null } A = \dim \ker(T - I) \leq 2k + 2, \quad \text{with equality iff } T = I,$$

T the monodromy of the order- $(2k + 2)$ recurrence.

Proof sketch. Row u_i of $Ax = 0$ contains one inner coordinate, $c_i y_i$ with $c_i \neq 0$, so y_i is a three-term expression in x_{i-1}, x_i, x_{i+1} . Substituting into row v_i yields a relation on $x_{i-k-1}, \dots, x_{i+k+1}$ whose extreme coefficients $-e_i b_{i+k}/c_{i+k}$ and $-e'_{i-k} b'_{i-k-1}/c_{i-k}$ are nonzero: a linear recurrence of order $2k + 2$. Its solutions on $\mathbb{Z}$ form a $(2k + 2)$ -space; T is the product of the n step matrices and $\ker A \cong \ker(T - I)$. $\square$

Rotation-invariant certificates. With constant weights $(a, 1, c, d, e)$ the blocks are $\begin{pmatrix} a+s_m & c \\ d+et_m & e \end{pmatrix}$, $s_m = 2 \cos \frac{2\pi m}{n}$, and null A counts grid values that are roots of the symbol

$$F(s) = (a + s)(d + et(s)) - c^2, \quad \deg F = k + 1.$$

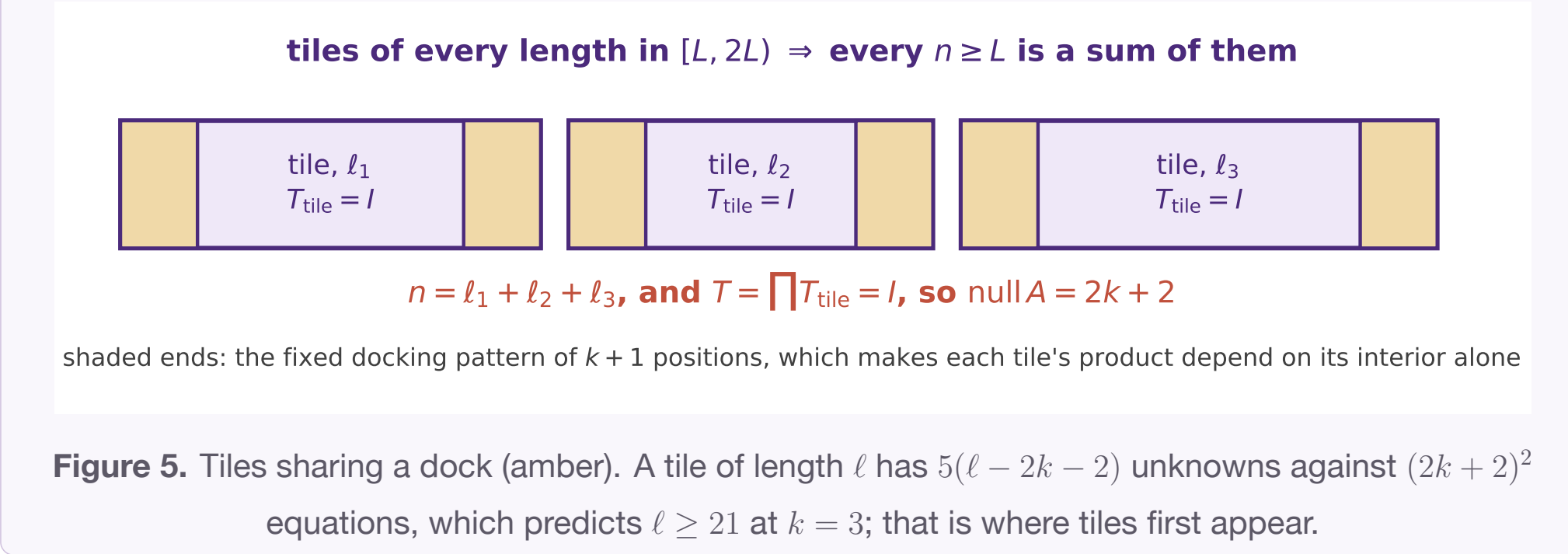
Three roots can be prescribed; for $k = 2$, $c^2 = -(s_1 + s_2)(s_1 + s_3)(s_2 + s_3)$, so antipodal roots are forbidden, which settles $k = 2$ for all $n \geq 9$ except $n = 10$ (done by an integer matrix). Galois orbits of $2 \cos \frac{2\pi}{d}$ give $Z(P(n, 4)) = 10$ for 60, 70, 90 $|n$ and $Z(P(n, 5)) = 12$ for $24 | n$.

Theorem 7 (rational-symbol certificates). A rotation-invariant certificate with rational symbol exists iff n is divisible by one of: 7, 9, 15, 20 ($k = 2$); 10, 24, 42 ($k = 3$); 60, 70, 90 ($k = 4$); 24, 126 ($k = 5$); none ($k = 6$); 120 ($k = 7$). Proof: a rational symbol has a Galois-closed root set, a union of orbits O_d of total size $k + 1$; the product of the Ψ_d must have the $k - 2$ forced middle coefficients and $c^2 \neq 0$; finitely many cases, enumerated exactly. Hence no rotation-invariant matrix reaches $2k + 2$ for all large n .

Breaking the symmetry: tiles. Fix a *dock* of $k + 1$ weight positions; a *tile* carries the dock at both ends and is free inside. Any window of width $2k + 2$ sees one tile's interior and dock entries only, so $T = \prod_{\text{tiles}} T_{\text{tile}}$ (Figure 5).

Theorem 8 (tiling). If identity tiles exist for every length in $[L, 2L]$, then $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for every $n \geq L$.

Proof sketch. $n < 2L$: one tile. $n \geq 2L$: peel a tile of length L and recurse. The monodromy of the concatenation is I ; apply Theorem 6 and the upper bound. $\square$



Existence of the tiles (computer-assisted). $T_{\text{tile}} = I$ iff the one-tile matrix on $P(\ell, k)$ has nullity $2k + 2$, i.e. $A(w)K(X) = 0$ with $K = \begin{pmatrix} c \\ x \end{pmatrix}$, a bilinear system with integer coefficients. For each required length a solution is shown to exist by Krawczyk's fixed-point theorem, with every bound evaluated in exact rational arithmetic ($\alpha < 1$ as an integer comparison): 77 tiles and 17 single- n instances, $\alpha \leq 1.0 \times 10^{-9}$. This is the only step not proved by hand.

Theorem 9 (main result of Part II).

$$Z(P(n, k)) = M(P(n, k)) = 2k + 2$$

for all $n \geq 17$ ($k = 3$), $n \geq 29$ ($k = 4$), $n \geq 162$ ($k = 5$), with no congruence condition. A finite case analysis (computer-checked, 1.2×10^9 candidate sets at $P(28, 4)$) settles the rest:

n	7	8	9	10	11	12	13	14	15	16	17
$Z(P(n, 3))$	6	6	6	8	7	7	8	8	8	8	8
$Z(P(n, 4))$	–	–	6	6	7	6	8	8	9	8	9

Corollary. Krishnan's Conjecture 5 holds with the optimal threshold 13; the $k = 4$ row is the first exact determination at any $k \geq 4$.

A withdrawn conjecture. An earlier version of this project conjectured that on a K_4 base the maximum nullity stays 6 while $Z = n + 2$, giving $Z - M \rightarrow \infty$ on cubic graphs. **This is false:** the three voltage-0 edges form a triangle, and a rank-one block uu^T on each lifted triangle gives rank $A \leq 3n$, so null $A \geq n$. The cover theorem requires $\det M(\zeta) \neq 0$, a hypothesis the earlier statement omitted. The replacement is a conjecture on the leading coefficient of $\det M(\zeta)$, proved for three families of bases and open in general.

Discussion

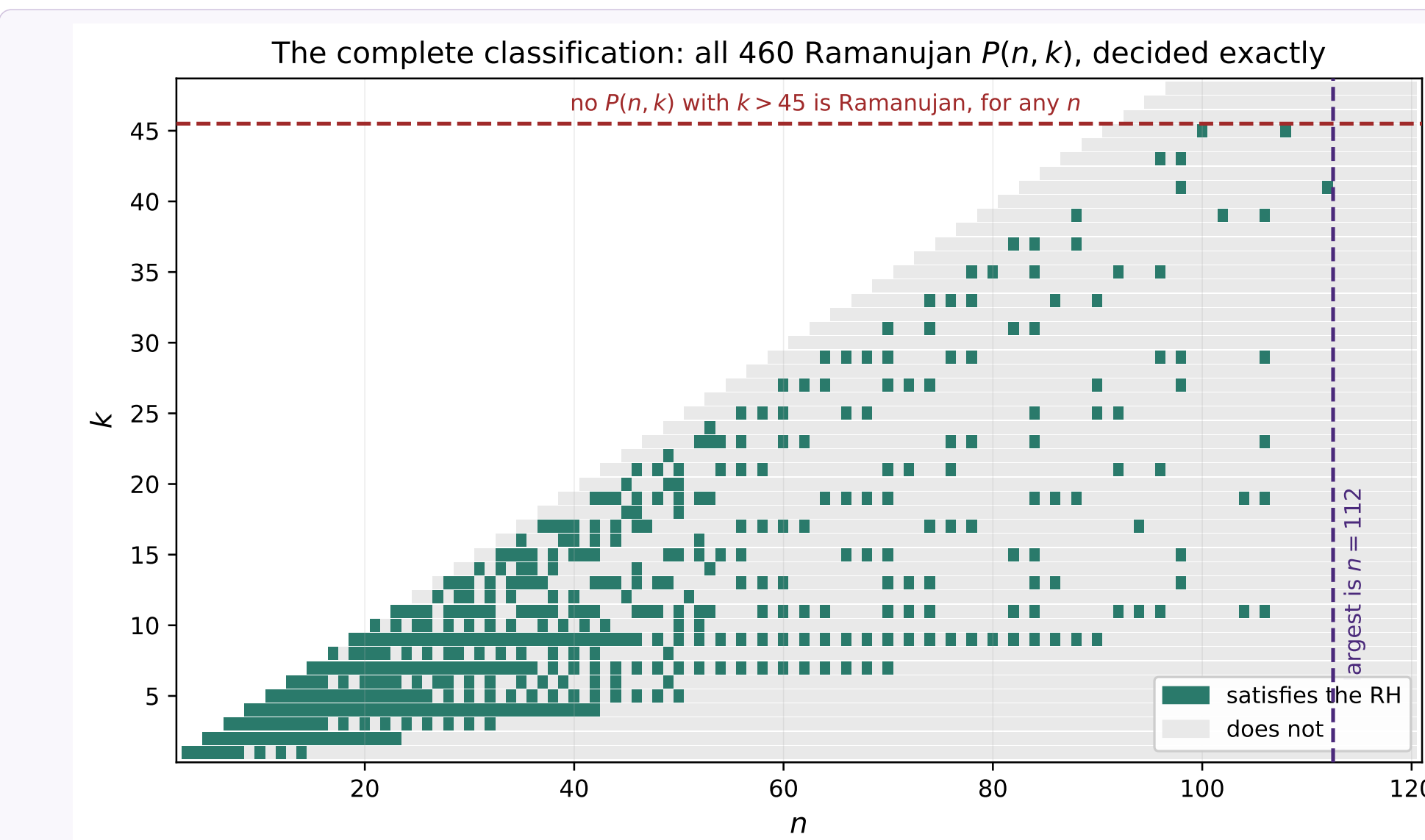


Figure 4. The answer to Q1: all 460 Ramanujan pairs. Unbroken runs where Theorem 5 applies; gaps where an interior band appears; nothing above $k = 45$.

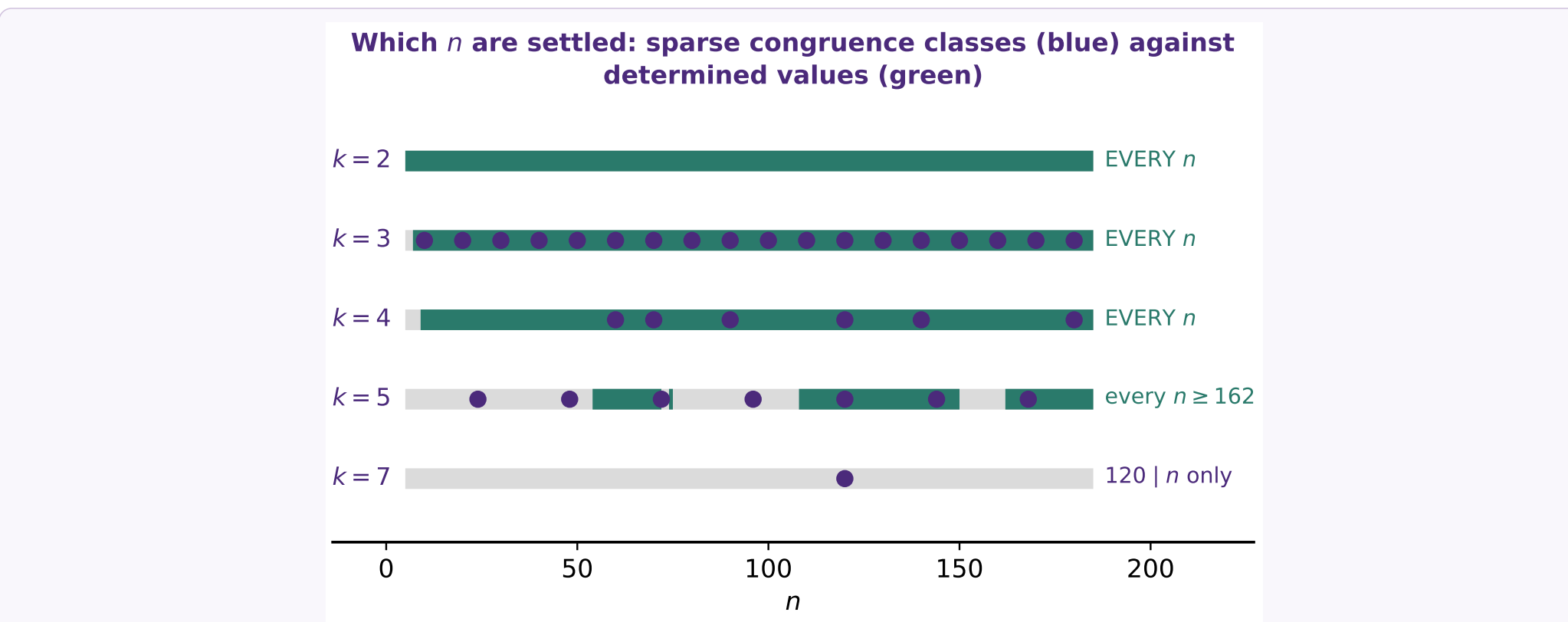


Figure 6. Which n are settled, by k : the sparse congruence classes reached by rotation-invariant certificates (dots) against the values determined for every n by tiles and exhaustive search (solid bars); at $k = 7$ only $120 | n$ is known.

Conclusions

- Q1.** Exactly 460 generalized Petersen graphs satisfy the Riemann Hypothesis. The criterion $Q \geq 0$ contains no k , the bound $n \leq 230$ holds for every k , and for $k \leq 9$ ($k \neq 6, 8$) the answer is the closed form of Theorem 5.
- Q2.** $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for all $n \geq 17, 29, 162$ at $k = 3, 4, 5$; Z is known at every n for $k \leq 4$; Krishnan's Conjecture 5 is proved with the optimal threshold 13.
- By Theorem 6 the matrix method reaches $2k + 2$ or nothing; by Theorem 7 cyclotomic certificates reach only finitely many congruence classes; tiles reach every large n . Every step is proved by hand except tile existence, which is a finite exact computation.
- For a cyclic cover of any fixed base, $M_1 \rightarrow M_0$ as $n \rightarrow \infty$ and M_0 has top eigenvalue 3, so no infinite Ramanujan family is a cyclic cover; on $P(n, k)$ the deficit $\lambda_2 - 2\sqrt{2}$ stays below $3 - 2\sqrt{2}$.
- Part I is a classification of a known type (cf. Le–Sander [6]); Part II closes two open problems from [4] and [5].

Open problems

- $M(P(24, 4))$: $Z = 10$ there, no nullity-10 matrix found, and no rotation-invariant one can exist.
- An effective threshold $L(k)$: identity tiles of every large length now exist for every k (an elliptic dock, a first-order root-space count, and a second-order bracket argument in $\mathfrak{sp}(2k + 2)$), so $Z = M = 2k + 2$ for all large n at every k , but the bound is ineffective and no tile beyond $k = 5$ has been exhibited.
- The leading-coefficient criterion for the cover ceiling: prove or refute.
- A proof of tile existence without computation, e.g. an algebraic construction of tiles from two cyclotomic certificates.

References and further reading

Cited. [1] A. Terras, *Zeta Functions of Graphs: A Stroll through the Garden*, Cambridge Univ. Press, 2010. [2] R. Gera, P. Stănică, The spectrum of generalized Petersen graphs, Australas. J. Combin. 49 (2011) 39–45. [3] AIM Minimum Rank Group, Zero forcing sets and the minimum rank of graphs, Linear Algebra Appl. 428 (2008) 1628–1648. [4] S. Rashidi, N. Shajareh Poursalavati, M. Tavakkoli, J. Algebra Combin. Discrete Struct. Appl. 7 (2020) 183–193. [5] A. Krishnan, A correction to the zero forcing number of $P(n, 3)$, arXiv:2607.19412 (2026). [6] A. Le, J. Sander, Integral circulant Ramanujan graphs of prime power order, Electron. J. Combin. 20 (2013). [7] R. Krawczyk, Newton–Algorithmen zur Bestimmung von Nullstellen mit Fehlerschranken, Computing 4 (1969) 187–201. **Background reading.** [8] S. Hoory, N. Linial, A. Wigderson, Expander graphs and their applications, Bull. AMS 43 (2006) – the survey to start with. [9] A. Lubotzky, R. Phillips, P. Sarnak, Ramanujan graphs, Combinatorica 8 (1988) – where the name comes from. [10] A. Nilli, On the second eigenvalue of a graph, Discrete Math. 91 (1991) – the two-page Alon–Boppana proof. [11] G. Davidoff, P. Sarnak, A. Valette, *Elementary Number Theory, Group Theory and Ramanujan Graphs*, Cambridge, 2003. [12] L. Hogben, J. Lin, B. Shader, *Inverse Problems and Zero Forcing for Graphs*, AMS, 2022 – the textbook for Part II. [13] S. Fallat, L. Hogben, The minimum rank of symmetric matrices described by a graph: a survey, Linear Algebra Appl. 426 (2007). [14] B. Brimkov, C. Fast, I. Hicks, Computational approaches for zero forcing and related problems, European J. Oper. Res. 273 (2019) – forts. [15] J. Gross, T. Tucker, *Topological Graph Theory*, Wiley, 1987 – voltage graphs and covers. [16] M. Watkins, A theorem on Tait colorings with an application to the generalized Petersen graphs, J. Combin. Theory 6 (1969). [17] R. Moore, R. Kearfoot, M. Cloud, *Introduction to Interval Analysis*, SIAM, 2009.

Every number on this board is regenerated from the stored certificates (147 checks); computation in Python (NumPy, SymPy, mpmath); all figures by the author. [AUTHOR: insert assistance/independence statement – see source.]